Short-term Load Forecasting

Power Systems Lab

Team members -

Tamajeet Biswas	2023EEB1247
Ansh Singh	2023EEB1186
Arjun Maindarkar	2023EEB1188

Contents

Why do we need Short-Term Load Forecasting ?

Short Term Load Forecasting is essential for

- ❑ Efficient power system operation
- ❑ Economic Dispatch & Cost Reduction
- ❑ Electricity Market Decisions
- ❑ Unit Commitment & Scheduling

A BRIEF INTRODUCTION TO THE PAPER

Two types of load forecasting models:

- Weather-based models
- Univariate models

Weather-based models, even though commonly used are less effective for very short-term horizons.

Univariate models:

- Use only past load data
- Suitable for short-term forecasting

Data Trends in Univariate Methods:

- Intraday (within a day)
- Intraweek (within a week)
- Annual (over a year)

Annual Seasonality is ignored because it has minimal impact on short term horizons.

LOAD SERIES DATA AND ITS PRE-PROCESSING

Data Used -

- 3 years of half-hourly electricity load data (British and French load series)
- First 2 years → model training and parameter optimization
- Final 1 year → out-of-sample testing for forecast accuracy

Preprocessing Steps

- **Log Transformation**
  - Applied natural logarithm to load values
  - Stabilizes variance and supports additive modeling structure
- **Handling Irregular Days (e.g., public holidays)**
  - Such days show abnormal load patterns not captured by models
  - Replaced using average of corresponding periods from adjacent weeks
- **Data Cleaning for Model Reliability**
  - Irregular days are excluded during:
    - parameter tuning
    - forecast accuracy evaluation

INTRODUCTION TO THE MODELS

- Paper evaluates five univariate methods:
  - Holt-Winters (HWT)
  - Intraday Cycle (IC)
  - Discount Weighted Regression (DWR)
  - DWR Spline
  - SVD-based method

- Then their Performance comparison is done with:
  - ARMA models
  - Artificial Neural Networks (ANN)
  - Weather-based models

HWT Exponential Smoothing

Extension of Holt-Winters Exponential Smoothing

Purpose is to handle data that has two seasonal patterns -

- Intraday cycle(patterns repeating after 24 hrs)
- Intraweek cycle(patterns repeating after every week)

Following is the state-space model of this method -

$$y_t = l_{t-1} + d_{t-m1} + w_{t-m2} + \phi e_{t-1} + \varepsilon_t$$

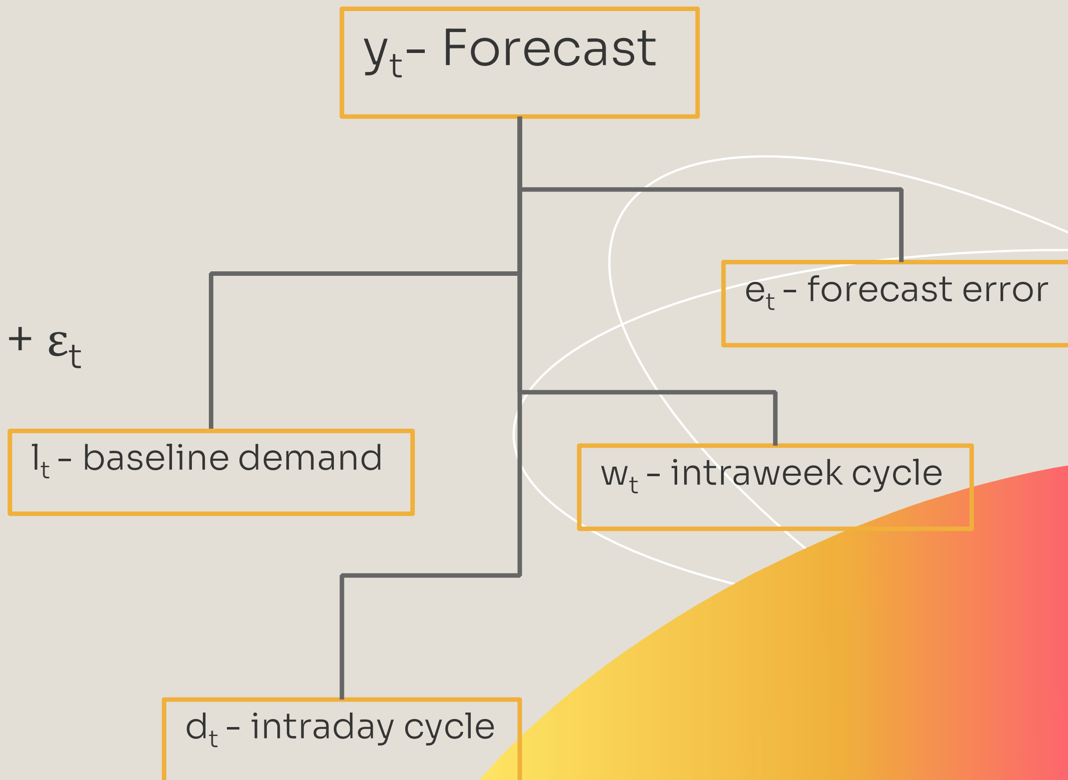
where -

$$e_t = y_t - (l_{t-1} + d_{t-m1} + w_{t-m2})$$

$$l_t = l_{t-1} + \alpha e_t$$

$$d_t = d_{t-m1} + \delta e_t$$

$$w_t = w_{t-m2} + \omega e_t$$



IC Exponential Smoothing

- IC Exponential smoothing method is similar to the Holt-Winters exponential smoothing method. The only difference is that there is no term for the intraweek cycle.
- An IC model was implemented with distinct intraday cycles for Monday, Friday, Saturday, and Sunday, and a common intraday cycle for the other days of the week.
- So, the only values used were: Level (l_t), Intraday cycle (d_t), Additive White Gaussian Noise (ε_t), Adjustment Term (ϕe_{t-1}).
- The analysis can be performed on French Load Consumption data. The data from year 2007 to end of 2009 has been used.
- The main reason why this method was used in earlier times was 1) It groups 'midweek' days and the other days with the same load pattern respectively. 2) We don't have to maintain a separate data for weekly load consumption. This reduces computation complexity.
- Restricted form: Some parameters

$$y_t = l_{t-1} + d_{t-m1} + \phi e_{t-1} + \varepsilon_t$$

DWR using Trigonometric terms

Basic Idea

- Uses a linear regression model with sine & cosine terms
- This models intraday seasonality (repeating daily patterns)

Problem it Solves

- Load patterns repeat daily, but their shape, magnitude, and timing evolve over time due to changing demand behavior and external influences.
- EWR uses a single decay factor, so it cannot adapt differently to changes in the level and seasonal components of the load.

How it Solves the Problem

Uses multiple discount factors:

$\lambda_1 \rightarrow$ intercept (level) $\lambda_2 \rightarrow$ main seasonal harmonics $\lambda_3 \rightarrow$ higher-order trigonometric terms

Continued...

This model is essentially a Fourier series regression model with time varying parameters

The summation of sine and cosine terms captures repeating patterns:

- Daily cycles (morning/evening peaks)
- Intraday variations

Each harmonic adds a new wave pattern, and combining many of them allows the model to closely match real-world load curves with peaks and dips.

The coefficients change over time and evolve at different rates using the discount factors specified.

$$y_t = \beta_{0,t} + \sum_{k=1}^{K} \left[\alpha_{k,t} \cos\left(\frac{2\pi k t}{S}\right) + \gamma_{k,t} \sin\left(\frac{2\pi k t}{S}\right) \right] + \varepsilon_t$$

where -

y_t : Load at time t

$\beta_{0,t}$: Time-varying level (intercept) $\rightarrow \lambda_1$

$\alpha_{k,t}, \gamma_{k,t}$: Seasonal coefficients (sine & cosine)

- Main harmonics $\rightarrow \lambda_2$
- Higher-order harmonics $\rightarrow \lambda_3$

S : Seasonal period (336 for weekly half-hourly data)

k : Harmonic index (multiple seasonal patterns)

ε_t : Random error

DWR Spline

Basic Idea – Dividing the week into segments with every point called a knot and a simple cubic equation is fitted into each segment

Problem – If ordinary least squares regression is used, we get one fixed spline for the entire history. But we know that the shape of the week changes over the year

Solution – Instead of OLS, we use DWR

What does this change?
When fitting the spline, the recent data is given more weight than older data which helps the model to update over time

This model uses different discount factors for different part of the week to allow the "baseline" to move differently from the "shape"

λ_1	Base knot	0.988	The level (baseline) changes at a moderate pace.
λ_2	Night-time knots	0.996	The night pattern is extremely stable.
λ_3	Other knots	0.995	The day pattern is very stable.

SVD-based Exponential Smoothing

Idea - To remove noise and reduce dimensionality in time-series data to make the forecast smoother and accurate.

Problems in real world data -

- **contains noise**
- **has daily/weekly patterns**
- **has redundancy**

Goals

1. **Denoise**
2. **Extract dominant patterns**
3. **Reduce complexity**

Load data Matrix with rows representing weeks and columns representing time slots(336 half-hrs for weekly data)

↓

Decomposing these 336 columns into K<336 columns which are uncorrelated

↓

Building a matrix P which shows how every intraweek feature vector has a corresponding interweek feature series

↓

Interpretation of SVD Output

Model flow -

Matrix representation of data

Single Value Decomposition

Interpretation

Rank-1 Decomposition

Truncated Representation

Projection onto Reduced space

Equations used in SVD-based exponential smoothing

$$y_t = p_{t-1}\,\tilde{V}'_{[t \bmod m_2]} + \phi e_{t-1} + \varepsilon_t$$ →

Observation Equation

- Here, p_{t-1} is a 1 x k state vector representing the estimated values of the first k interweek feature series at the previous step.
- **V**: An (m2 x m2) matrix whose columns are orthogonal basis functions called **intraweek feature vectors.** $V_{[t \bmod m2]}'$ is the transpose of the row that corresponds to the current period of the week.
- ϕe_{t-1} is the residual auto correction term.

$$e_t = y_t - p_{t-1}\,\tilde{V}'_{[t \bmod m_2]}$$

→ Error Calculation

- This calculates the prediction error at time t by comparing the actual load to the projected state before the autocorrelation adjustment.

$$p_t = p_{t-1} + \left(\alpha \mathbf{1}_{m_2} \tilde{V} + \delta \sum_{j=1}^{7} \tilde{V}_{[t \bmod m_1]+(j-1)m_1} + \omega \tilde{V}_{[t \bmod m_2]}\right) e_t$$ → State update Equation

- It updates the state vector for the next iteration. The update is dependent on the error we calculated above.

Benchmark Methods

ARMA - AutoRegressive Moving Average

A classic statistical time series model

Core Idea - Predict future load using a combination of -

1. Past load values(AutoRegressive part)
2. Past forecast values(Moving average part)

To handle the daily and weekly cycles,model uses three “lag” terms -

1. Regular recent hours
2. The same time yesterday(L^{m1})
3. The same time last week(L^{m2})

Box-Jenkins Method used

ANN - Artificial Neural Network

A machine learning model inspired by brain

Core Idea - A “feedforward” method with one hidden layer is trained to map past loads to a future load.

Flow -

- Data Prep: To make the data easier for the network to learn, they first removed the daily and weekly cycles by applying a seasonal difference $(1-L_{m1})(1-L_{m2})$.
- One Model per Horizon: Instead of one model predicting all future times, we build a separate ANN for each lead time
- Inputs: For a model predicting "h" hours ahead, the inputs include the load at the current time and at specific lags (e.g., 1 hour ago, yesterday at this time, last week at this time).
- Tuning: We use a hold-out sample (last 6 months of data) to choose the best settings: which inputs to use, how many neurons in the hidden layer, and the learning rate.

EMPIRICAL COMPARISON OF UNIVARIATE METHODS

- The main metric we have used to evaluate the accuracy of these models is Mean Absolute Percentage Error (MAPE). The formula is given below:

$$\text{MAPE} = \frac{1}{n} \sum \left| \frac{\text{actual value} - \text{forecast value}}{\text{actual value}} \right|$$

- The other metrics, such as mean absolute error, root mean squared percentage error, and root mean squared error did not have much difference when comparing between the different methods.
- As we will see the graphs in the next slide, we will notice that the MAPE increases as we increase the hours horizon. The reason for this is that as horizon increases, previous data becomes less relevant (autocorrelation degradation), and hence the errors in further forecasts accumulate, resulting in a higher MAPE.
- The MAPE is still not too high because of the auto correction term we have introduced (ϕe_{t-1}).
- We will now analyze the graphs and compare the performances of various methods in the next slide.

Comparing results from graphs

Here we can see that the new SVD-based exp smoothing performs better than the Total & split exp, DWR with trig terms, DWR spline and spline based exp smoothing methods.

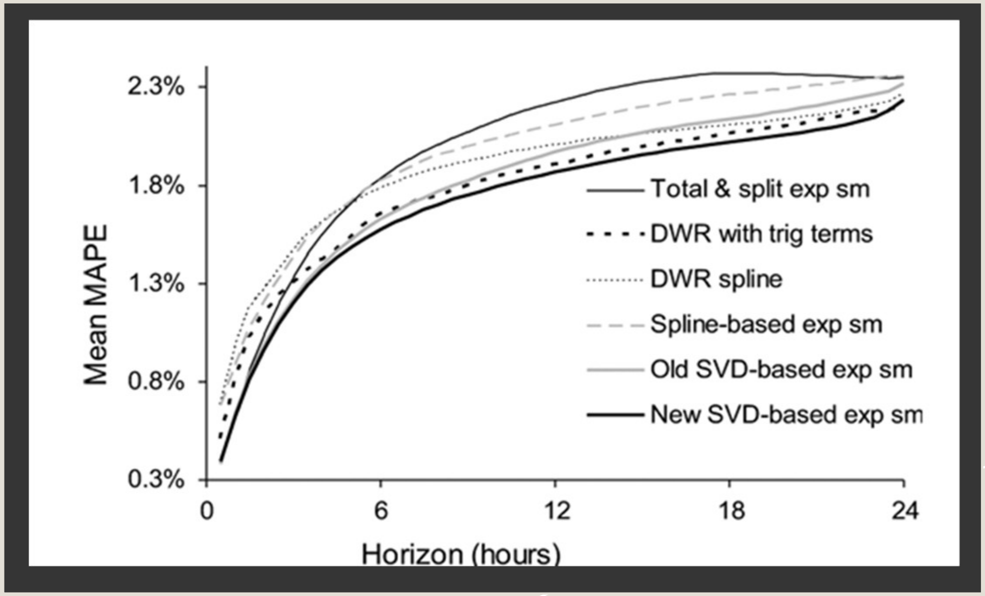


Fig. 1. Mean MAPE for the first 4 methods and the new SVD-based method

EMPIRICAL COMPARISON OF UNIVARIATE METHODS WITH WEATHER BASED MODELS

- Our objective now is to evaluate the forecast accuracy of our univariate methods with weather based models.
- The same metric of Mean Absolute Percentage Error (MAPE) is used to evaluate the model forecasting accuracy.

Weather-Based Method

- Regression models are fitted at exactly 10 selected points on the daily load curve.
- Inputs:
  - Forecasted temperature
  - Relative Humidity
  - Wind Speed
- Intermediate load values:
  - Obtained via heuristic interpolation using past load profiles

Comparing results from graphs

We can easily observe from the graph results obtained that :-

- Exponential smoothing methods outperform weather-based models for short horizons (upto 6 hours)

- Weather-based models perform better for longer horizons (> 5 hours)

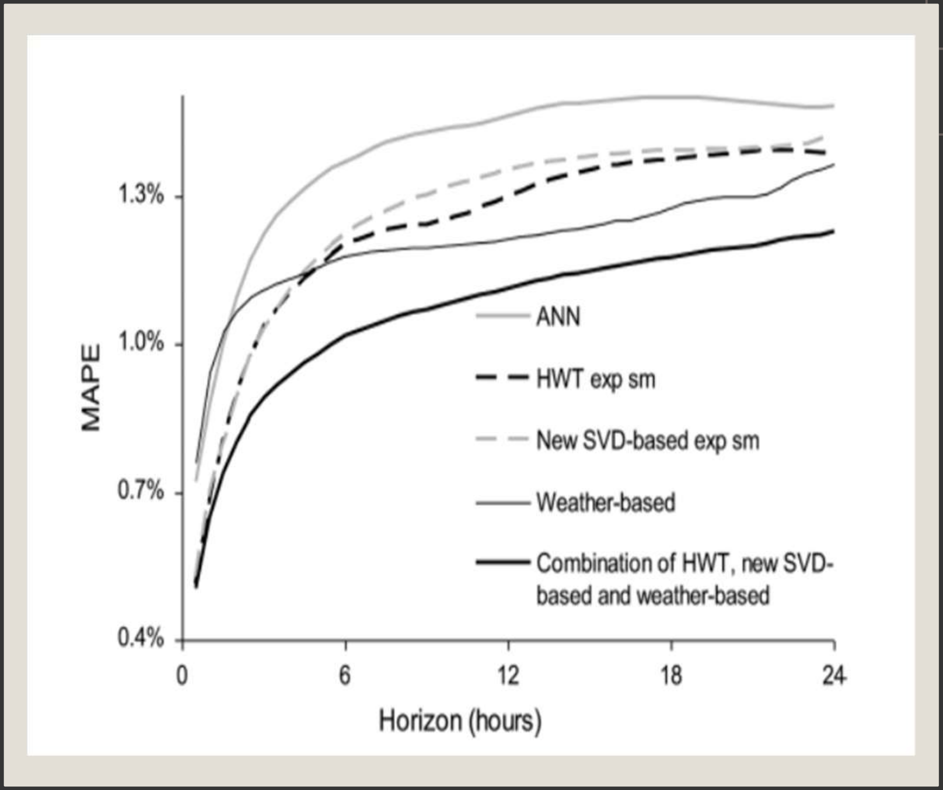


Fig. 2. MAPE comparison with benchmark methods

Updates after midterm work

- We have used the following specifications in our implementation:
  - MATLAB
  - Python
  - UK Load Dataset from 31st March, 2022 to 1st April, 2023. This data is taken in log values of load in MW.
  - IEEE compatible output figures (Graphs are captured in 600 dpi resolution to maintain quality).
  - Python (For implementing the weather based forecasting model)

- Data Pre-processing:
  - The load data was obtained from structured datasets, and the relevant demand variable was extracted for analysis.
  - Missing, invalid, and non-numeric entries were removed, and data continuity was ensured through interpolation where necessary.
  - The time series was organized into a uniform half-hourly resolution, maintaining consistent temporal indexing.
  - A logarithmic transformation was applied to stabilize variance and improve model robustness.
  - The dataset was partitioned into training and testing sets in chronological order.
  - Intraday and intraweek seasonal characteristics were retained during preprocessing to support accurate forecasting.
  - The processed data was structured to ensure compatibility across multiple forecasting models for consistent performance evaluation.

Plots - HWT Model

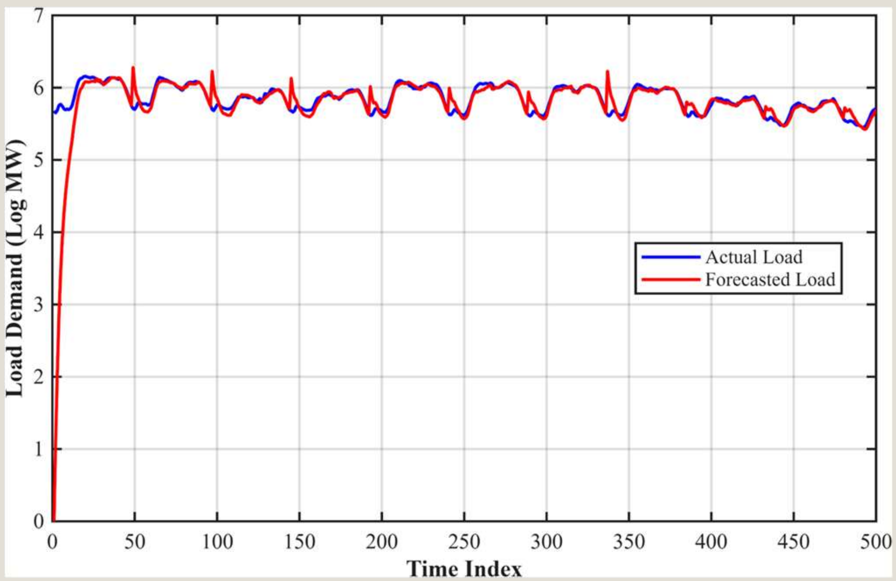


Fig. 3. Comparison of actual and forecasted load demand using the HWT model.

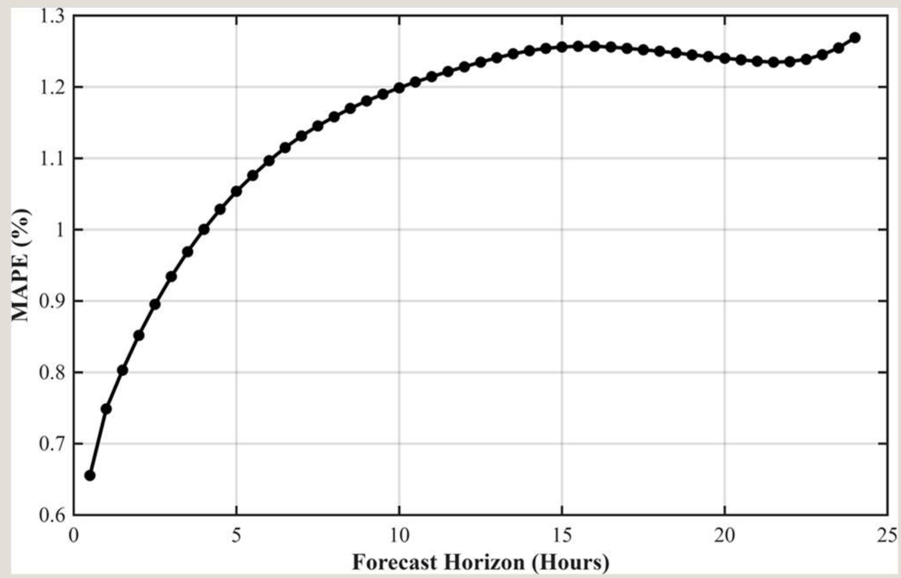


Fig. 4. MAPE versus forecasting horizon for the Holt-Winters-Taylor (HWT) model.

Plots - IC Exponential Model

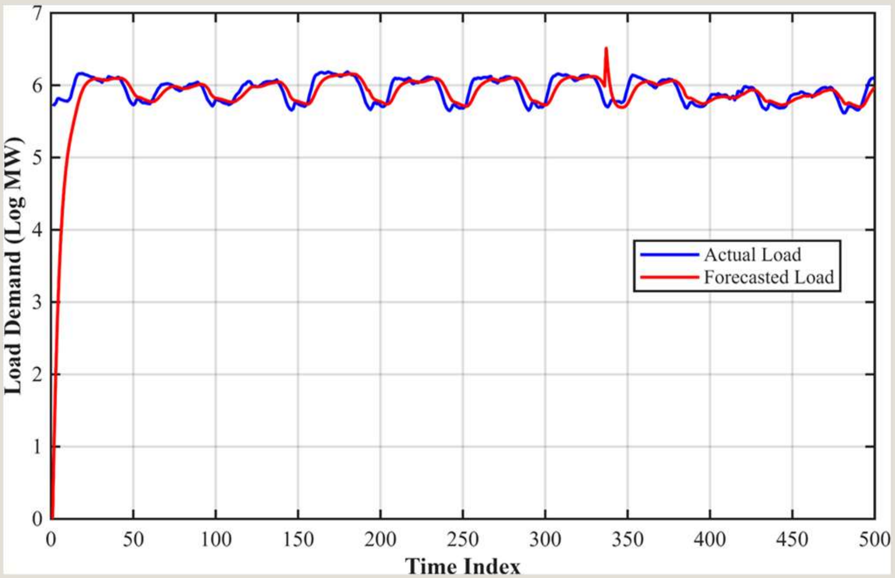


Fig. 5. Comparison of actual and forecasted load demand using the IC model

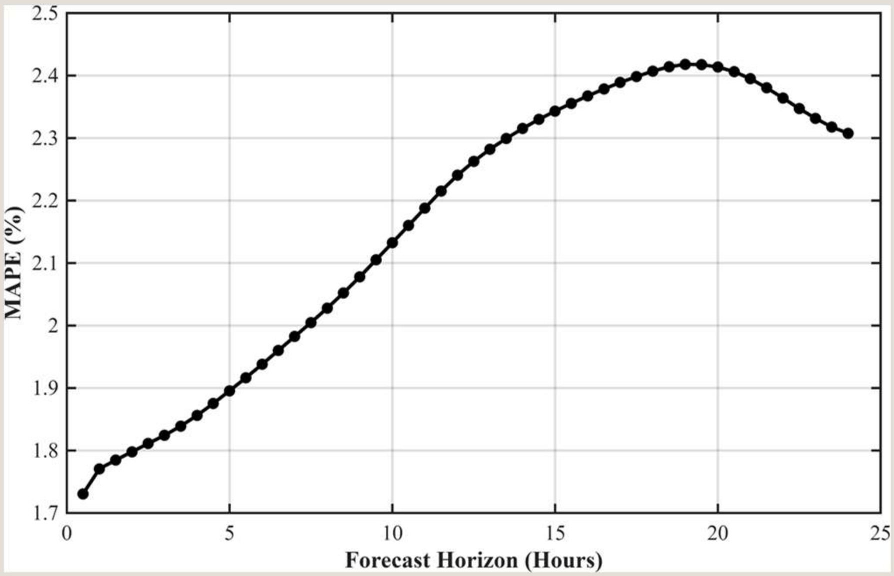


Fig. 6. MAPE versus forecasting horizon for the Intraday Cycle (IC) model.

Plots - DWR with Trigonometric terms

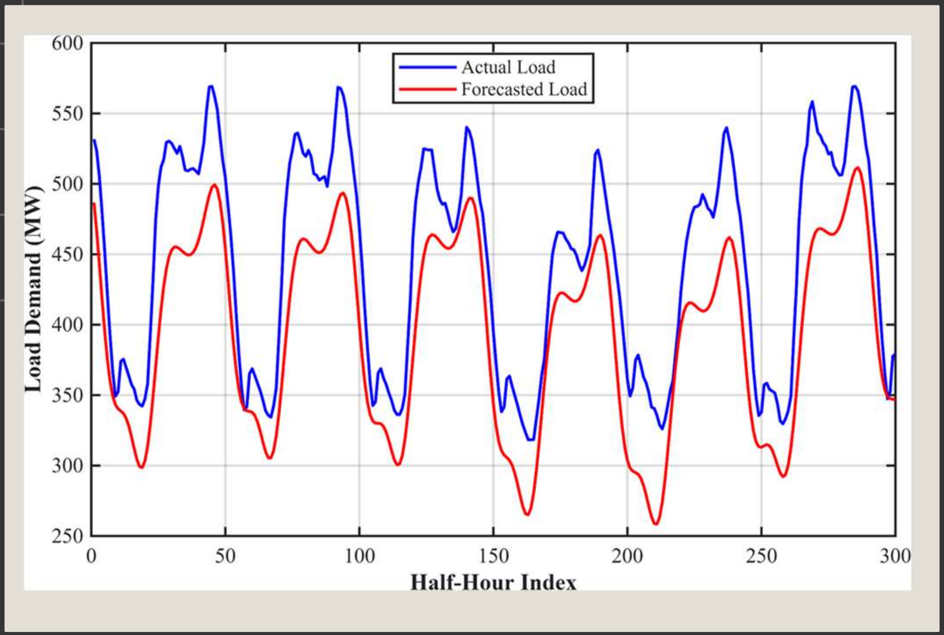


Fig. 7. Comparison of actual and forecasted load demand using the DWR Trigonometric model.

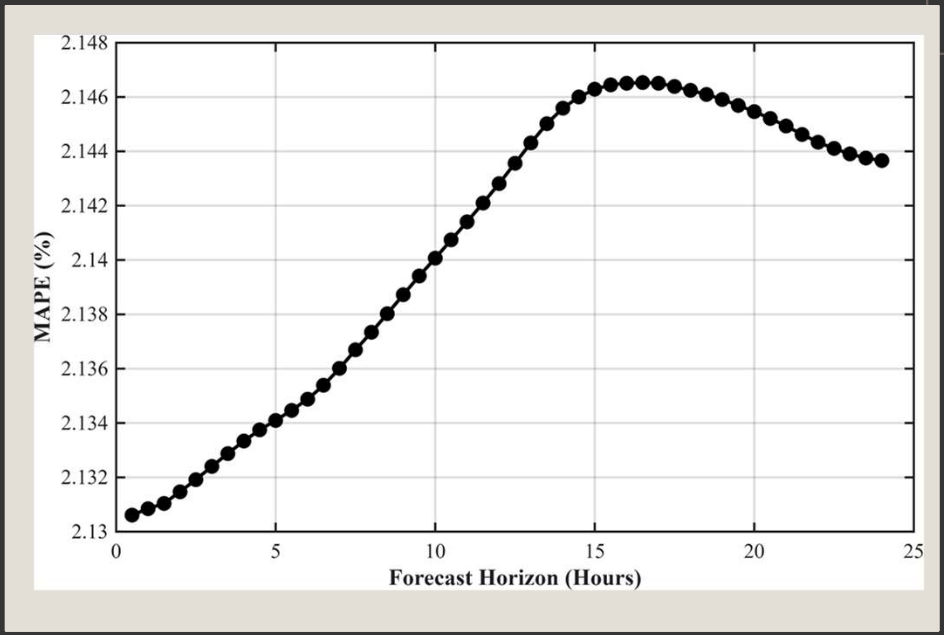


Fig. 8. MAPE versus forecasting horizon for the DWR Trigonometric model.

Plots - DWR Spline

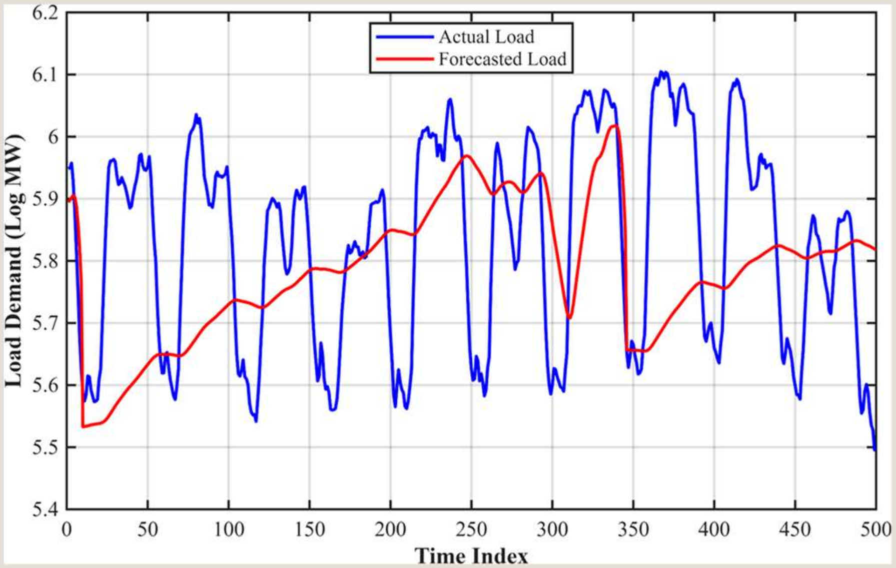


Fig. 9. Comparison of actual and forecasted load demand using the DWR Spline model.

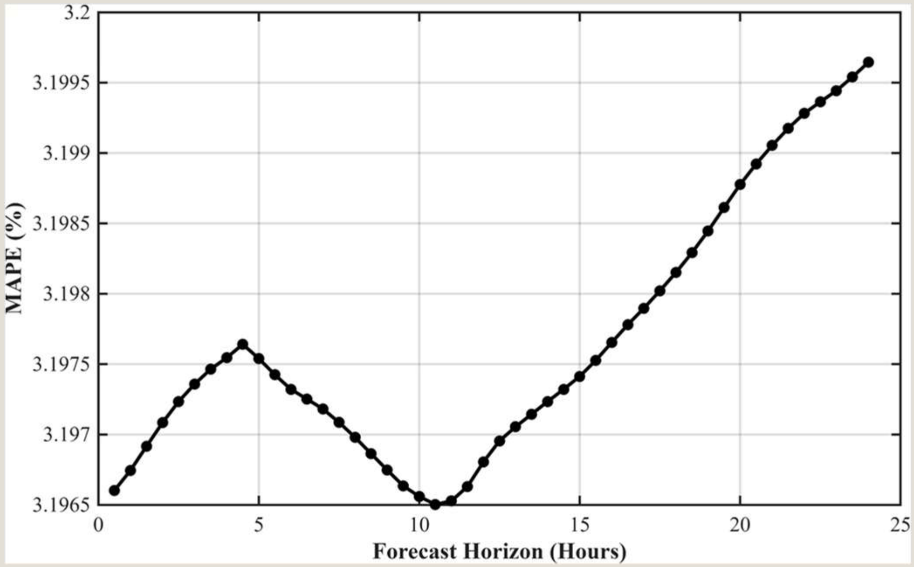


Fig. 10. MAPE versus forecasting horizon for the DWR Spline model.

Plots - SVD

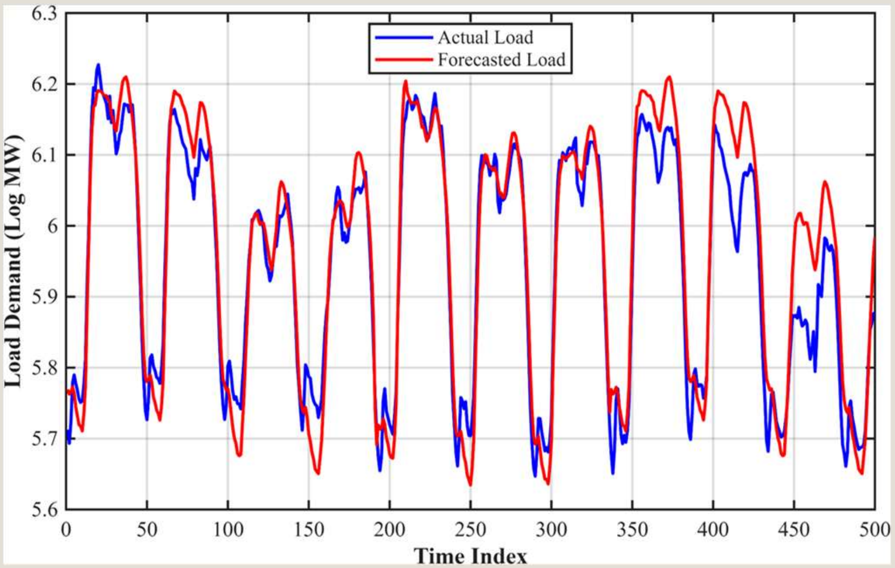


Fig. 11. Comparison of actual and forecasted load demand using the DWR Spline model.

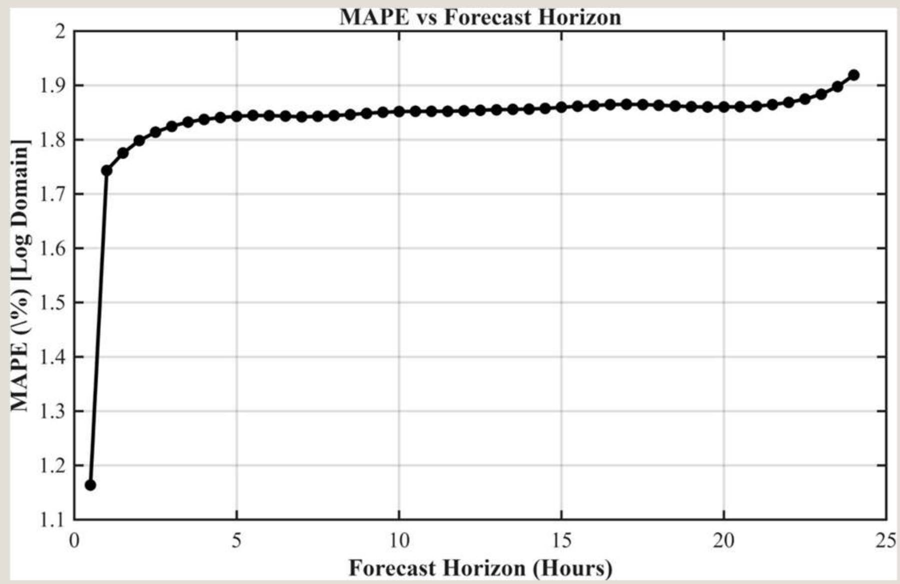


Fig. 12. MAPE versus forecasting horizon for the SVD-based exponential smoothing model.

Plots - ARMA

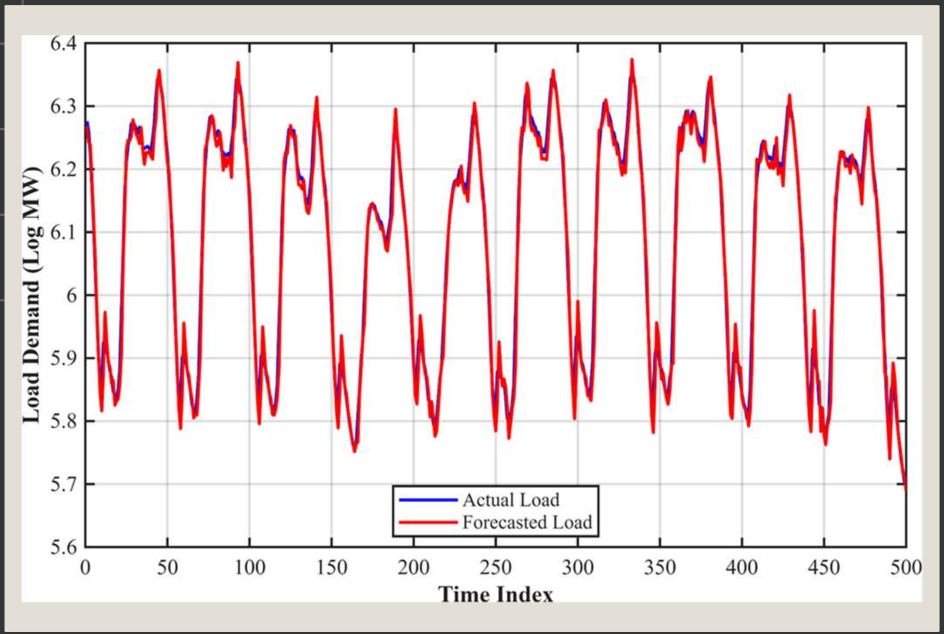


Fig. 13. Comparison of actual and forecasted load demand using the ARMA model.

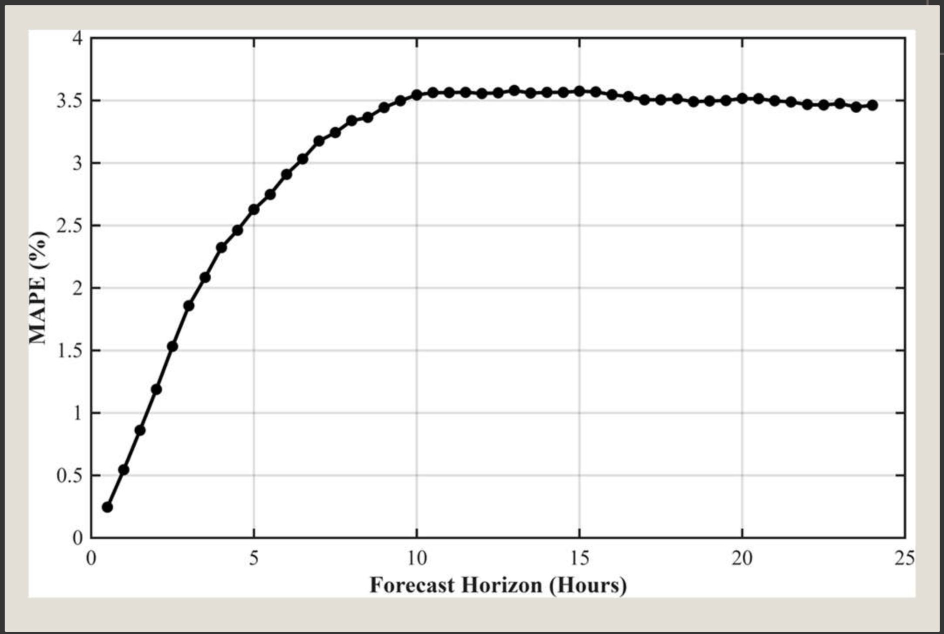


Fig. 14. MAPE versus forecasting horizon for the ARMA model.

Plots - ANN

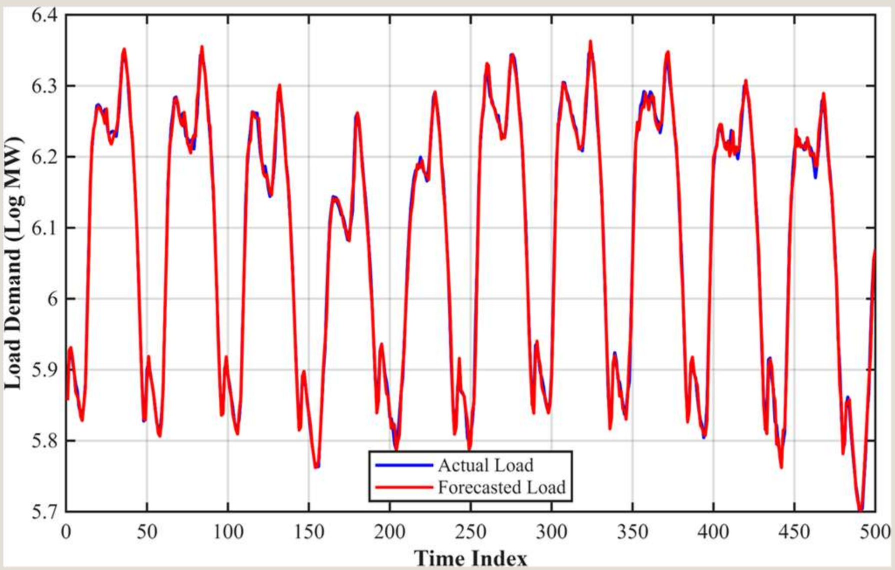


Fig. 15. Comparison of actual and forecasted load demand using the ANN model.

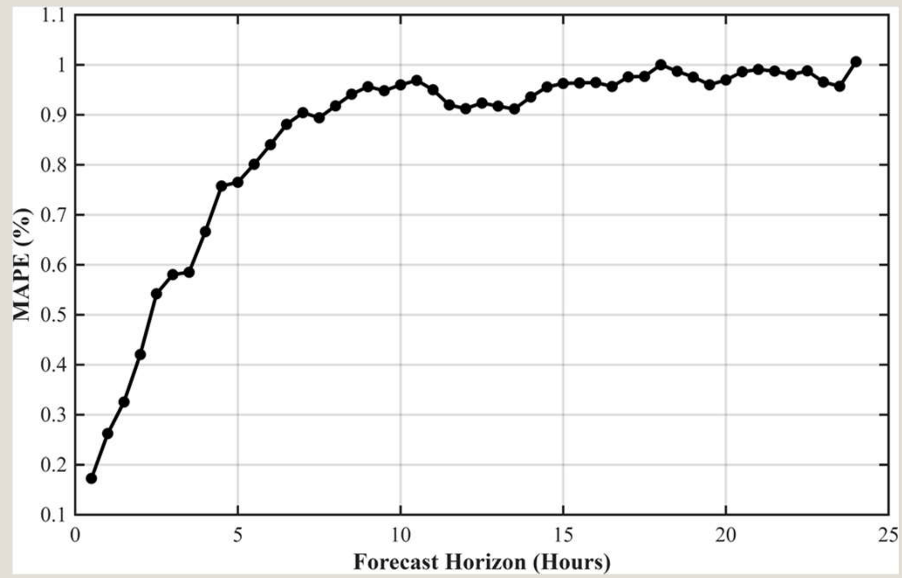


Fig. 16. MAPE versus forecasting horizon for the ANN model.

Weather based Model

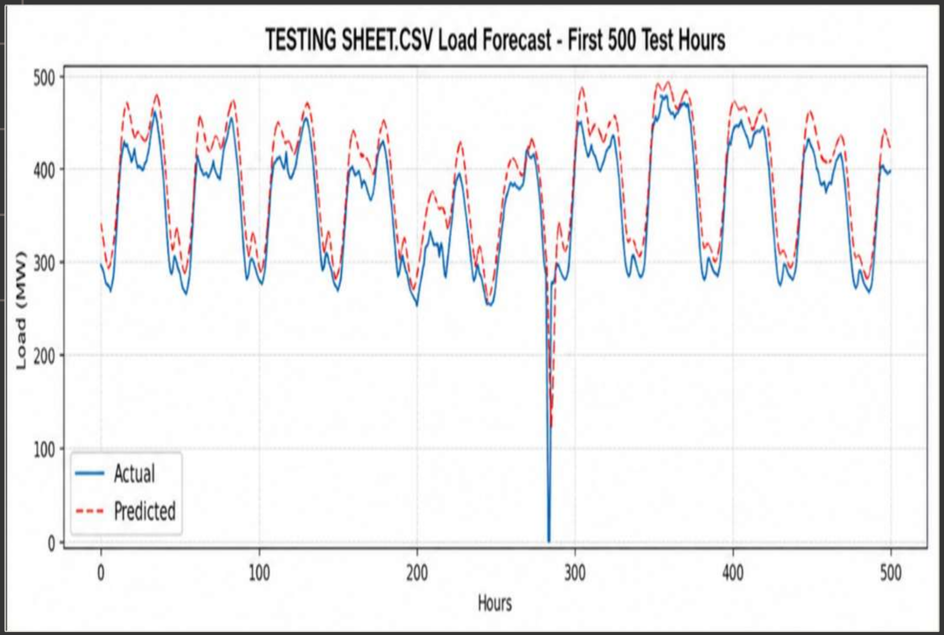


Fig. 17. Comparison of actual and forecasted load demand using the Weather based LSTM model

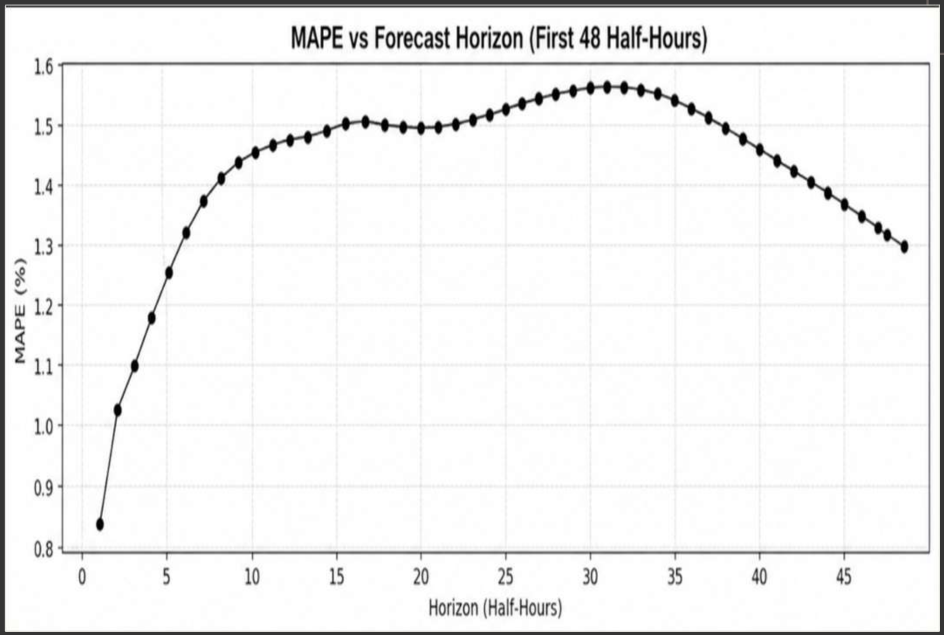


Fig. 18. MAPE versus forecasting horizon for the Weather based LSTM model

Comparison Plot

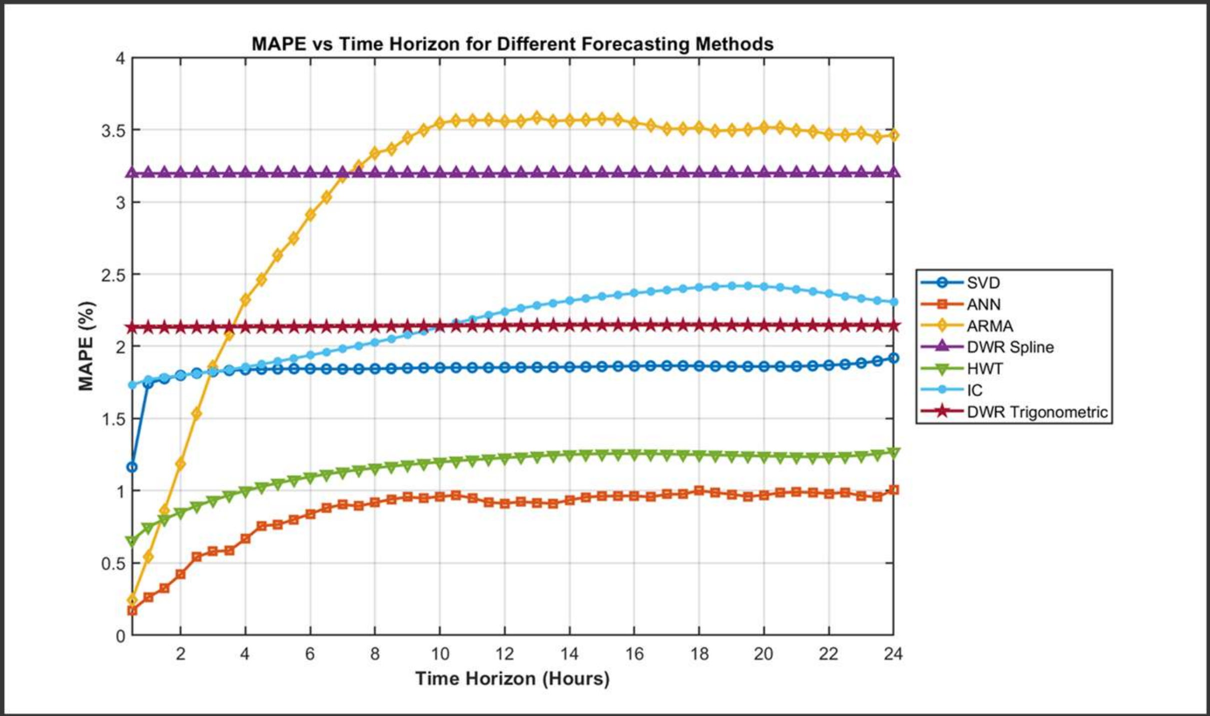


Fig. 19. Comparison of MAPE versus forecasting horizon for all implemented forecasting models.

Conclusion

Key Findings

- Evaluated 5 exponentially weighted methods for short-term load forecasting and compared them to benchmark methods like ARMA,ANN and also weather based forecasting
- Although ANN gave the best performance,Univariate methods gave significantly close and accurate results.
- Among the 5 univariate methods, HWT performed the best followed closely by SVD-based Exponential Smoothing and IC. DWR Spline and Trigonometric also gave a satisfactory performance but poorer than the other methods.
- ARMA gave a significantly good result for shorter time horizon but its MAPE increased as we increased the time horizon of investigation.

Performance Insights

- Best univariate methods outperform weather-based models for short horizons (up to 6 hours)
- For longer horizons : combining univariate + weather-based methods gives best results

Comparison with Existing Methods

- HWT and IC methods perform strongly
- HWT preferred for:
  - Simplicity
  - Ability to generate prediction intervals

Limitations

- Prediction intervals for DWR-based models are not straightforward
- Some newer methods did not outperform traditional benchmarks

Future Scope

- Apply SVD with weather-based models
- Extend methods to electricity price forecasting

Thank you

Find our Github Project on this link:
https://github.com/programmingmaster69/Power-Systems-EE309-Load-Forecasting